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FPGA Design Using VHDL

Module 6G: User-Defined Packages

User-Defined Packages

- A VHDL package includes a set of related and/or reusable declarations
- A package can be shared across multiple design units by using the “use” clause
- Declaration section of a package contains declaration statement for all elements in the package
 - For some elements, the declaration is all that is needed
 - For other elements, a functional description is also needed
- Packages can be standardized or user-defined

Packages – Function Example

- package 'bit_pack': One function to add two 4-bit numbers

```
library ieee;
use ieee.std_logic_1164.all;

package bit_pack is
    function add4(
        add1      : std_logic_vector (3 downto 0);
        add2      : std_logic_vector (3 downto 0);
        carry     : std_logic)
    return std_logic_vector;
    -- more functions defined here if you want. Will be part of bit_pack
end package bit_pack;

package body bit_pack is
    function add4(
        add1      : std_logic_vector (3 downto 0);
        add2      : std_logic_vector (3 downto 0);
        carry     : std_logic)
    return std_logic_vector is
    variable cout, cin: std_logic;
    variable ret_val : std_logic_vector (4 downto 0);

    begin
        cin:= carry;
        ret_val:="00000" ;
        for i in 0 to 3 loop
            ret_val(i) := add1(i) xor add2(i) xor cin;
            cout:= (add1(i) and add2(i)) or (add1(i) and cin) or (add2(i) and cin);
            cin:= cout;
        end loop;
        ret_val(4):=cout;
        return ret_val;
    end add4;
end package body;
-- save this file as bit_pack.vhd
```

Package declaration

Package body

Packages - Example

```
library ieee;
use ieee.std_logic_1164.all;
use work.bit_pack.all;

-- addfour.vhd
entity addfour is
port ( a: in STD_LOGIC_VECTOR (3 downto 0);
      b: in STD_LOGIC_VECTOR (3 downto 0);
      cin: in STD_LOGIC;
      sum: out STD_LOGIC_VECTOR (4 downto 0) );
end addfour;

architecture behav of addfour is

begin
    sum <= add4(a,b,cin); -- not quite the same as signal
                        -- binding
end addfour;
```

Package Example – no body

- Typically used to “globally” define constants that can be used through the hierarchy of the design

```
library ieee;
use ieee.std_logic_1164.all;
package external is
-- Memory access size type.
subtype memsize_t is std_logic_vector(1 downto 0);
    constant MEM_B8: memsize_t := "00";
    constant MEM_B16: memsize_t := "01";
    constant MEM_B32: memsize_t := "10";
end;
-- save file as external.vhd
```

- Include this package in every level of hierarchy that uses the constants defined in the package:
 - o use work.external.all;